H02 · S03–S04 (Devore 2.3–2.5) — Worked Solutions

Source: Probability and Statistics Homework H02 · study-slides (vetame21.github.io/study-slides)

> [ADDED] The original homework has no solutions. These worked solutions were written for this site and every number was checked in code. Compare them with your lecturer's model answers when those are released.

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1 Product rule (adapted from Devore Ex. 2.37). 2 temperatures, 3 pressures, 4 catalysts; each run uses one of each.

(a) How many different runs are possible?

> **Answer:** 24

$2 \times 3 \times 4 = 24.$

Why this method: Product rule: one choice per stage.

(b) How many use the lower temperature and one of the two lowest pressures?

> **Answer:** 8

$1 \times 2 \times 4 = 8.$

Why this method: Product rule with restricted stages.

(c) One run is chosen at random. P(it uses the lower temperature)?

> **Answer:** $1/2$

$\frac{1\times3\times4}{24} = \frac{12}{24} = \frac{1}{2}.$

Why this method: Equally likely outcomes: $N(A)/N.$

2 Permutations or combinations? (adapted from Devore Ex. 2.30). 12 bottles from different wineries: 5 zinfandel, 4 merlot, 3 cabernet.

(a) Serve 3 of the zinfandels one after another, order matters. How many ways?

> **Answer:** 60 (permutation)

Order matters: $P_{3,5} = 5 \cdot 4 \cdot 3 = 60$.

Why this method: Ordered selection → permutation.

(b) Choose 3 of the 12 bottles, order ignored. How many selections?

> **Answer:** 220 (combination)

$\binom{12}{3} = \frac{12 \cdot 11 \cdot 10}{3!} = 220$.

Why this method: Unordered selection → combination.

(c) How many selections in (b) have one bottle of each variety?

> **Answer:** 60

$\binom{5}{1}\binom{4}{1}\binom{3}{1} = 5 \cdot 4 \cdot 3 = 60$.

Why this method: Product of type-by-type choices.

(d) P(one of each variety)?

> **Answer:** $60/220 = 3/11 \approx 0.273$

$\frac{60}{220} = \frac{3}{11}$.

Why this method: $N(A)/N$ with matching (unordered) counts.

(e) P(all three the same variety)?

> **Answer:** $15/220 = 3/44 \approx 0.068$

Disjoint cases: $\binom{5}{3} + \binom{4}{3} + \binom{3}{3} = 10 + 4 + 1 = 15$, so $\frac{15}{220} = \frac{3}{44}$.

Why this method: Split into disjoint cases and add.

3 Counting the votes (adapted from Devore Ex. 2.36). Six slips: four A, two B, counted one at a time in random order (every ordering of AAAABB equally likely).

(a) How many different orderings are there?

> **Answer:** 15

Choose the 2 positions for the B's: $\binom{6}{2} = 15$.

Why this method: Combination: only which positions hold B matters.

(b) List the orderings in which A is strictly ahead after every slip.

> **Answer:** AAAABB, AAABAB, AAABBA, AABAAB, AABABA

The first two slips must both be A (after 2 slips A must lead, so AB fails). Then check each remaining ordering prefix by prefix; e.g. AABBAA fails at slip 4 (2–2).

Why this method: Enumerate, checking the condition at every prefix.

(c) P(A stays strictly ahead throughout)?

> **Answer:** $5/15 = 1/3$

$\frac{5}{15} = \frac{1}{3}$. (This matches the ballot theorem: $\frac{a-b}{a+b} = \frac{4-2}{6} = \frac{1}{3}$.)

Why this method: $N(A)/N$.

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4 Conditional probability from a table (adapted from Devore Ex. 2.45). Percent of population by ethnic group and blood type — Group 1: O 8, A 10, B 1, AB 1; Group 2: 14, 14, 1, 1; Group 3: 22, 16, 8, 4. A = type A, B = type B, C = group 3.

(a) Find $P(A), P(C), P(A \cap C)$.

> **Answer:** $P(A) = 0.40,\ P(C) = 0.50,\ P(A \cap C) = 0.16$

Type A column: $10 + 14 + 16 = 40\%$. Group 3 row: $22 + 16 + 8 + 4 = 50\%$. Both: 16%.

Why this method: Read margins and cells.

(b) Find $P(A \mid C)$ and $P(C \mid A)$; explain each and why they differ.

> **Answer:** $P(A \mid C) = 0.32,\ P(C \mid A) = 0.40$

$P(A \mid C) = \frac{0.16}{0.50} = 0.32$: among group 3, 32% are type A. $P(C \mid A) = \frac{0.16}{0.40} = 0.40$: among type A people, 40% are in group 3. Same overlap, different reference groups.

Why this method: S04: the condition sets the denominator.

(c) If the person does NOT have type B, P(group 1)?

> **Answer:** $19/90 \approx 0.211$

$P(B) = 1 + 1 + 8 = 10\%$, so $P(B') = 0.90$. Group 1 without type B: $8 + 10 + 1 = 19\%$.
$P(\text{G1} \mid B') = \frac{0.19}{0.90} = 0.211$.

Why this method: Restrict to the condition, then divide.

5 Total probability and Bayes (adapted from Devore Ex. 2.59). Regular 50%, plus 30%, premium 20%; of these, 30%, 60%, 50% fill their tanks (F).

(a) Draw a tree with every branch probability.

> **Answer:** First split 0.5 / 0.3 / 0.2; then F | each: 0.3, 0.6, 0.5 (and F′: 0.7, 0.4, 0.5).

Check each split sums to 1.

Why this method: S04: probability tree.

(b) P(plus and fills).

> **Answer:** 0.18

$P(A_2 \cap F) = 0.3 \times 0.6 = 0.18.$

Why this method: Multiply along one route.

(c) P(fills).

> **Answer:** 0.43

$0.5(0.3) + 0.3(0.6) + 0.2(0.5) = 0.15 + 0.18 + 0.10 = 0.43.$

Why this method: Total probability: add all routes to F.

(d) Given fills, P(regular), P(plus), P(premium); check they sum to 1.

> **Answer:** $0.349,\ 0.419,\ 0.233$ (sum 1)

Bayes: divide each route by $P(F) = 0.43$: $\frac{0.15}{0.43} = 0.349$, $\frac{0.18}{0.43} = 0.419$, $\frac{0.10}{0.43} = 0.233$. Sum $= 1$ ✓.

Why this method: Bayes' rule = one route ÷ all routes to the observation.

6 A witness at night (adapted from Devore Ex. 2.62). Blue Cab runs 20% of taxis, Green 80%. At night a witness identifies colour correctly 80% of the time, whatever the colour. The witness says the taxi was blue.

(a) Identify the prior and the two likelihoods.

> **Answer:** Prior $P(\text{blue}) = 0.2$; $P(\text{says blue} \mid \text{blue}) = 0.8$; $P(\text{says blue} \mid \text{green}) = 0.2$

Why this method: Name before you calculate.

(b) Total P(says blue).

> **Answer:** 0.32

$0.2(0.8) + 0.8(0.2) = 0.16 + 0.16 = 0.32.$

Why this method: Total probability.

(c) P(blue | says blue).

> **Answer:** 0.5

$\frac{0.16}{0.32} = 0.5.$

Why this method: Bayes' rule.

(d) The witness is right 80% of the time. Why is the posterior so much lower?

> **Answer:** Base rate: green cabs are 4× as common.

False 'blue' reports from the many green cabs (0.16) are as frequent as true 'blue' reports from the few blue cabs (0.16), so a 'blue' report is only 50/50.

Why this method: S04: likelihood ≠ posterior.

7 Independent projects (adapted from Devore Ex. 2.71). A = Asian project succeeds, B = European project succeeds, independent, P(A) = 0.5, P(B) = 0.6.

(a) If the Asian project fails, P(the European one also fails)? Explain.

> **Answer:** 0.4

Complements of independent events are independent, so $P(B' \mid A') = P(B') = 1 - 0.6 = 0.4$.

Why this method: Independence: the condition changes nothing.

(b) P(at least one succeeds).

> **Answer:** 0.8

$1 - P(A' \cap B') = 1 - (0.5)(0.4) = 0.8$.

Why this method: Complement of 'both fail'.

(c) Given at least one succeeds, P(only the Asian one succeeds).

> **Answer:** 0.25

$P(A \cap B') = 0.5 \times 0.4 = 0.2$, and it lies inside $A \cup B$, so $\frac{0.2}{0.8} = 0.25$.

Why this method: Conditional probability: restrict to the union.

(d) Are A and B mutually exclusive?

> **Answer:** No: $P(A \cap B) = 0.3 \neq 0$

Independence gives $P(A \cap B) = (0.5)(0.6) = 0.3 > 0$, so both can succeed together.

Why this method: S04: independent ≠ mutually exclusive.

8 Relief valves (adapted from Devore Ex. 2.78). Three valves, each opens with probability 0.9, independently.

(a) P(all three open).

> **Answer:** 0.729

$0.9^3 = 0.729.$

Why this method: Independence → multiply.

(b) P(at least one fails to open).

> **Answer:** 0.271

$1 - 0.729 = 0.271.$

Why this method: Complement of 'all open'.

(c) P(at least one opens).

> **Answer:** 0.999

$1 - 0.1^3 = 0.999.$

Why this method: Complement of 'all fail'.

(d) Why is the complement rule efficient in (b) and (c)?

> **Answer:** Each 'at least one' event has many routes; its complement is a single route.

'At least one fails' covers 7 of the 8 outcomes; its opposite is just one: all open. One multiplication instead of seven terms.

Why this method: S04: reliability via complements.